District Vote Aggregation Bridge

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Draft — June 28, 2026

Abstract

District vote aggregation is an optional bridge from verified RCOUNT count summaries to RPLAN district assignments. The bridge does not make RCOUNT a district-plan validator. Instead, it composes independently owned artifacts: the RCOUNT package content hash, the RPLAN plan hash, the optional RCTX context hash, and the optional RCTX crosswalk hash used to project reporting units into plan units. The implemented `rcount-district` crate verifies the count package, validates the RPLAN plan, checks any supplied RCTX context against the plan unit universe, matches declared RCOUNT RCTX references, rejects crosswalk hash drift, and refuses non-integral weighted vote allocations. The result is a district aggregation transcript whose fields can be checked without collapsing RCOUNT, RPLAN, and RCTX into one authority.

1 Introduction

RCOUNT count packages and RPLAN district plans answer different questions. RCOUNT verifies count summaries and their reconciliation evidence. RPLAN records a district assignment and owns the plan hash for that assignment. RCTX records shared context and crosswalk identity. A district vote table needs all three claims, but it should not blur them.

The V.8 bridge is therefore optional. It takes an RCOUNT package whose summaries already verify, an RPLAN plan whose assignment validates under `rplan-core`, and, when supplied, an RCTX context or crosswalk. It emits a transcript that binds the RCOUNT package content hash, the RPLAN plan hash, and the relevant RCTX hashes before reporting district totals.

This paper covers that bridge. It does not claim that RCOUNT validates a plan, that RPLAN validates count evidence, or that RCTX decides which assignment is lawful. It claims only that the implemented bridge composes those artifacts and that the district totals follow from the verified summaries, the assignment, and any explicit crosswalk rows used for projection.

2 Bridge Contract

The bridge contract is a transcript over artifact identities and district totals. Its core fields are implemented by `DistrictAggregationTranscript` in `rcount-district`.

Field	Meaning
`aggregation_version`	Version string for the bridge transcript. Current value: `0.1-draft`.
`rcount_package_content_hash`	Content hash of the verified RCOUNT package.
`rplan_plan_hash`	Hash of the RPLAN district plan and assignment, computed by `rplan-core`.
`rctx_context_hash`	Optional RCTX context hash, present when a context is supplied and matches the plan unit universe.
`rctx_reference_id`	Optional RCOUNT reference id whose context hash matches the supplied context.
`rctx_crosswalk_hash`	Optional RCTX crosswalk hash, either from a matching RCOUNT reference or from an explicit crosswalk file.
`contest_id`	Contest whose summaries are aggregated.
`status`	Count status, such as canvassed, used to select the summaries.
`unit_universe_hash`	Unit universe hash from the RPLAN plan unit index.
`district_totals`	One RCOUNT-style summary per district, with source reporting unit ids.
`checks`	Per-district `district_aggregation_total` checks.

Without an explicit crosswalk, the bridge treats each RPLAN unit id as an RCOUNT reporting unit id and sums the matching summaries into the assigned district. With an explicit crosswalk, it reads RCTX crosswalk NDJSON rows, verifies the set with `rctx-core`, and projects source reporting units to plan units before summing by district. The crosswalk path is accepted only with a supplied RCTX context, so the projection has a context identity rather than a loose file name.

RCOUNT references are used only as consumer bindings. The core package verifier checks `rctx_refs` through `verify_rctx_references`: reference ids must be unique, context and crosswalk hashes must be SHA-256-shaped, and the role must be `unit-context`, `aggregation-crosswalk`, or `plan-context`. The bridge then matches the supplied context hash to a declared reference, preferring the aggregation-crosswalk role.

3 Implemented Surface

The public entry point is `aggregate_package_dir_with_plan_path`. It reads an RCOUNT package directory, an RPLAN document, an optional RCTX context, and an optional explicit crosswalk path, then calls `aggregate_package_districts`. The latter is the bridge function: it runs `verify_package`, validates the plan, checks that any supplied context validates and has the same unit universe as the plan, computes the package and plan hashes, resolves a matching RCOUNT RCTX reference, and emits the transcript fields.

Aggregation has two paths. `aggregate_direct` indexes summaries whose reporting unit ids are also plan unit ids. `aggregate_with_crosswalk` indexes all matching summaries, reads `rctx-core` crosswalk rows, maps each row's target unit to the RPLAN district assignment, and accumulates weighted totals. The weighted path refuses to round: `weighted_i64` raises `NonIntegralCrosswalkAllocation` if votes, undervotes, overvotes, blank contests, or counted ballots cannot be allocated integrally.

Check	Failure caught
`verify_package`	RCOUNT package failures before district aggregation, including invalid declared RCTX references.
`plan.validate`	Invalid RPLAN plan structure or assignment before any count totals are summed.
`validate_context_matches_plan`	RCTX context hash drift or a context unit universe that does not match the plan.
`validate_crosswalk_path`	Explicit crosswalk without context, declared crosswalk hash drift, or RCTX crosswalk verification failure.
`aggregate_direct`	Missing or duplicate plan-unit summaries for the requested contest and status.
`aggregate_with_crosswalk`	Missing source summaries and non-integral weighted allocations.
`verify_lineage_conservation`	Broken synthetic multi-election split/merge lineage before per-cycle district aggregation.

The multi-election helper `verify_synthetic_multi_election_harness` uses the same bridge after package verification, jurisdiction-total verification, and lineage conservation checks. Its cycle transcript records the package content hash, the RPLAN plan hash, current reporting units, lineage event count, district count, and pass status for each synthetic cycle.

4 Fixtures

The basic positive fixture is exercised by `aggregates_summary_basic_into_rplan_districts`. It builds the synthetic summary package, pairs it with `synthetic_summary_basic_rplan_document`, and verifies that the package summaries aggregate into two RPLAN districts.

The declared-reference fixture is `aggregation_transcript_records_matching_rctx_reference`. It adds an RCOUNT `rctx_refs` entry whose context hash matches the supplied context. The transcript records both `rctx_reference_id` and `rctx_crosswalk_hash`.

The explicit-crosswalk path is covered by `aggregation_validates_explicit_crosswalk_hash` and `aggregation_uses_explicit_crosswalk_projection`. The first test writes identity RCTX crosswalk rows, computes their set hash, and requires the transcript to carry that hash. The second writes rows that project both synthetic source precincts to one plan unit, proving that district totals are driven by the explicit crosswalk rather than by direct id equality.

The RCTX fixture path is `aggregation_consumes_minimal_rctx_fixture_crosswalk`. It consumes the minimal RCTX package fixture through stable context and crosswalk hashes and records the matching RCTX reference in the district transcript.

The crosswalk-hash negative fixture is `aggregation_rejects_declared_crosswalk_hash_drift`. It supplies an explicit crosswalk file while the RCOUNT reference declares a different crosswalk hash. The bridge rejects the aggregation with `CrosswalkHashMismatch`.

The multi-election positive fixture is `verifies_multi_election_harness_across_lineage_and_distri` It verifies the synthetic 2024, 2026, and 2028 cycles, including a split lineage event and a merge lineage event, before aggregating each cycle by district.

The stale-plan-unit negative fixture is `rejects_multi_election_harness_with_stale_plan_unit`. The 2028 plan names a unit that has no current matching summary, and the bridge fails with `MissingPlanUnitSummary`. The broken-lineage negative fixture is `rejects_multi_election_harness_with` It edits the current unit in the 2028 merge lineage and fails before aggregation under the RCOUNT lineage-conservation check.

5 Boundaries

The claim boundary is the one stated in the V.8 plan. RCOUNT owns count summaries and reconciliation. RPLAN owns assignments and plan hashes. `rplan-core` validates the plan and computes the assignment-bearing plan hash. RCTX owns context and crosswalk identity; `rctx-core` verifies the crosswalk record set and computes its hash. The bridge composes these hashes and sums counts. It does not make RCOUNT a plan-validity verifier.

The implemented crosswalk path is intentionally conservative. It can use explicit RCTX crosswalk rows and reject declared hash drift, but the transcript does not yet record non-exhaustive crosswalk caveats. Rows whose target unit is not in the plan are ignored by the current aggregation path, so a future public fixture should make the non-exhaustive boundary visible rather than treating it as a silent policy choice.

Fractional allocation is also bounded. The bridge accepts rational crosswalk weights only when every affected count field allocates to an integer. That is appropriate for a count transcript that emits integer RCOUNT summaries, but it is not a general fractional redistricting report. A rational-output transcript mode remains future work for genuinely fractional population, area, or vote allocation records.

Finally, the implemented fixtures are synthetic. They prove artifact binding, lineage failure, stale-plan failure, explicit-crosswalk projection, and crosswalk-hash drift. They do not yet prove a real public precinct-to-district crosswalk ingestion path.

6 Conclusion

V.8 adds a narrow district vote aggregation bridge. The implemented `rcount-district` crate verifies the RCOUNT package, validates the RPLAN plan, optionally binds an RCTX context and crosswalk, and emits a transcript that carries the RCOUNT package content hash, RPLAN plan hash, optional RCTX context hash, optional RCTX reference id, optional crosswalk hash, district totals, and per-district aggregation checks.

That surface is deliberately modest. It composes public identities and count summaries; it does not transfer plan validation into RCOUNT or count validation into RPLAN. With explicit crosswalk projection and integral-allocation rejection in place, the remaining work is to add realistic public crosswalk fixtures, record non-exhaustive caveats, and introduce rational output where integer district summaries are the wrong target.

References

Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.

Gio Della-Libera. Rplan and rctx reproducible district plan artifacts, 2026b. Project documentation and Rust crates.